GreenTech Manufacturing

New Plant Investment Analysis

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Introduction

Land	Construction	Equipment	Total investment
$300K	**$500K**	**$200K**	**$1,000,000**

10-year Excel model in four connected sheets

Inputs
Case data + Assumptions

→

Decisions
Scenario switch

→

Calculations
Cash Flows

→

Outputs
NPV · IRR · Payback

Key assumptions: 10% discount rate · land sold at cost in Year 10 · COGS = labor + utilities

KEY FINDING 1

The Project Creates Value

Net Present Value

$2,867,892

Positive at a 10% discount rate

Internal Rate of Return

44.5%

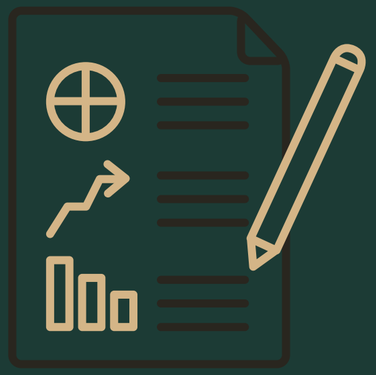
More than 4× the required return

Payback Period

2.72 years

Investment recovered in Year 3

Answer: GreenTech should proceed with the new plant.



KEY FINDING 2

Scenarios and Sensitivity

Scenario results

Scenario	NPV	IRR	Payback
Worst	$1,176,492	32.5%	3.18 yrs
Base	$2,867,892	44.5%	2.72 yrs
Best	$4,485,671	51.5%	2.54 yrs

NPV stays positive in every scenario.

Impact on base NPV

Driver	Low	High
Revenue growth	-34%	+44%
Discount rate	+16%	-14%
Labor cost	+4%	-4%

Revenue growth is the most sensitive driver.

Worst Case Outcome

EXCEL MODEL

Decisions				
Scenario selected	**Worst**			
Driver	Worst	Base	Best	Active
Revenue growth	3%	10%	15%	3%
Labor (Year 1)	$ 230,000	$ 200,000	$ 190,000	$ 230,000
Discount rate	12%	10%	9%	12%

Growth 3% · Labor $230K · Discount rate 12%

NPV $1,176,492 · IRR 32.5% · Payback 3.18 yrs

OUTPUTS		
Discount rate		12%
Net Present Value (NPV)	$	1,176,492
Internal Rate of Return (IRR)		32.5%
Payback period (years)		3.18

Recommendations

01 Proceed with the investment

02 Finance a significant portion with fixed-rate debt

03 Confirm material costs and plant capacity

04 Secure long-term customer contracts

05 Update the model annually with actual results

DECISION

Build the new plant

Positive NPV in all three scenarios

